

```
1 --s16_1b
2
3 LIBRARY ieee;
4 USE ieee.std_logic_1164.ALL;
5 USE ieee.numeric_std.ALL;
6
7 ENTITY s16_1 IS
8     PORT (
9         clk : IN STD_LOGIC;
10        s : IN STD_LOGIC;
11        xy : OUT STD_LOGIC_VECTOR(1 DOWNTO 0)
12    );
13END ENTITY;
14
15ARCHITECTURE behavioral OF s16_1 IS
16    SIGNAL q : STD_LOGIC_VECTOR(3 DOWNTO 0) := x"4";
17
18BEGIN
19
20    -- sequential process
21    seq_p : PROCESS (clk)
22    BEGIN
23        IF rising_edge(clk) THEN
24            IF s = '1' THEN
25                q <= q(2 DOWNTO 0) & q(3);
26            ELSE
27                q <= q(0) & q(3 DOWNTO 1);
28            END IF;
29        END IF;
30    END PROCESS;
31
32    -- output logic implementation
33    xy <= q(1 DOWNTO 0);
34END ARCHITECTURE behavioral;
```